

Лабораторная работа №3

Компьютерный практикум по статистическому анализу данных

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Основной целью работы является знакомство с NETEM — инструментом для тестирования производительности приложений в виртуальной сети, а также получение навыков проведения интерактивного и воспроизводимого экспериментов по измерению задержки и её дрожания (jitter) в моделируемой сети в среде Mininet.

Задание

1. Задайте простейшую топологию, состоящую из двух хостов и коммутатора с назначенной по умолчанию mininet сетью 10.0.0.0/8.
2. Проведите интерактивные эксперименты по добавлению/изменению задержки, джиттера, значения корреляции для джиттера и задержки, распределения времени задержки в эмулируемой глобальной сети.
3. Реализуйте воспроизводимый эксперимент по заданию значения задержки в эмулируемой глобальной сети. Постройте график.
4. Самостоятельно реализуйте воспроизводимые эксперименты по изменению задержки, джиттера, значения корреляции для джиттера и задержки, распределения времени задержки в эмулируемой глобальной сети. Постройте графики.

Выполнение лабораторной работы

```
c0p0 none
mininet@mininet-vm:~$ sudo mn --topo=single,2 -x
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1)
*** Configuring hosts
h1 h2
*** Running terms on localhost:10.0
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> 
```

Рисунок 1: Простейшая топология

ifconfig

```
*host: h1*@mininet-vm
root@mininet-vm:/home/mininet# ifconfig
h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.1 netmask 255.0.0.0 broadcast 10.255.255.255
    ether e6:f9:a8:b4:81:62 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1052 bytes 273728 (273.7 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1052 bytes 273728 (273.7 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet#
```

```
*host: h2*@mininet-vm
root@mininet-vm:/home/mininet# ifconfig
h2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.2 netmask 255.0.0.0 broadcast 10.255.255.255
    ether de:3d:aa:f7:15:bc txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1282 bytes 290668 (290.6 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1282 bytes 290668 (290.6 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet#
```

Проверка подключения

```
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 6
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=5.02 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.293 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.111 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.064 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.066 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.118 ms

--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5078ms
root@mininet-vm:~# cat /dev/null > /dev/null
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
loop txqueuelen 1000 (Local Loopback)
RX packets 1282 bytes 290668 (290.6 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 1282 bytes 290668 (290.6 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions

root@mininet-vm:/home/mininet# ping 10.0.0.1 -c 6
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=2.60 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0.051 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=0.069 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0.093 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=0.060 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=0.061 ms

--- 10.0.0.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5096ms
rtt min/avg/max/mdev = 0.051/0.489/2.604/0.945 ms
root@mininet-vm:/home/mininet#
```

Добавление/изменение задержки в эмулируемой глобальной сети

```
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem delay 100ms
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 6
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=102 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=101 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=100 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=100 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=102 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=101 ms

--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5010ms
rtt min/avg/max/mdev = 100.404/101.004/102.023/0.611 ms
root@mininet-vm:/home/mininet#
```

Рисунок 4: Добавление задержки в 100мс

Добавление/изменение задержки в эмулируемой глобальной сети

```
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 6
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=202 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=202 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=201 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=201 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=201 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=202 ms

--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5011ms
rtt min/avg/max/mdev = 200.668/201.654/202.485/0.769 ms
root@mininet-vm:/home/mininet#
```

```
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1282 bytes 290668 (290.6 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1282 bytes 290668 (290.6 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
root@mininet-vm:/home/mininet# ping 10.0.0.1 -c 6
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=2.60 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0.051 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=0.069 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0.093 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=0.060 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=0.061 ms
```

```
--- 10.0.0.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5096ms
rtt min/avg/max/mdev = 0.051/0.489/2.604/0.945 ms
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h2-eth0 root netem delay 1
00ms
root@mininet-vm:/home/mininet#
```

Изменение задержки в эмулируемой глобальной сети

```
root@mininet-vm:/home/mininet# sudo tc qdisc change dev h1-eth0 root netem delay 50ms
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 6
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=103 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=102 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=101 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=102 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=101 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=101 ms

--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5012ms
rtt min/avg/max/mdev = 100.990/101.579/102.512/0.529 ms
root@mininet-vm:/home/mininet#
root@mininet-vm:/home/mininet# ifconfig h1-eth0 netmask 255.0.0.0
loop txqueuelen 1000 (Local Loopback)
RX packets 1282 bytes 290668 (290.6 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 1282 bytes 290668 (290.6 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet# ping 10.0.0.1 -c 6
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data:
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=2.60 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0.051 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=0.069 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0.093 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=0.060 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=0.061 ms

--- 10.0.0.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5096ms
rtt min/avg/max/mdev = 0.051/0.489/2.604/0.945 ms
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h2-eth0 root netem delay 100ms
root@mininet-vm:/home/mininet# sudo tc qdisc change dev h2-eth0 root netem delay 50ms
root@mininet-vm:/home/mininet#
```

Восстановление исходных значений (удаление правил) задержки в эмулируемой глобальной сети

```
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root netem
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 6
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.161 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.099 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.055 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.095 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.100 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.127 ms

--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5105ms
rtt min/avg/max/mdev = 0.055/0.106/0.161/0.032 ms
root@mininet-vm:/home/mininet# cat /sys/class/net/h1-eth0/ethtool
ethtool h1-eth0 (Local Loopback)
RX packets 1282 bytes 290668 (290.6 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 1282 bytes 290668 (290.6 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet# ping 10.0.0.1 -c 6
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=2.60 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0.051 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=0.069 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0.093 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=0.060 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=0.061 ms

--- 10.0.0.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5096ms
rtt min/avg/max/mdev = 0.051/0.489/2.604/0.945 ms
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h2-eth0 root netem delay 1
00ms
root@mininet-vm:/home/mininet# sudo tc qdisc change dev h2-eth0 root netem dela
```

Добавление значения дрожания задержки в интерфейс подключения к эмулируемой глобальной сети

```
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem delay 100ms 10ms
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 6
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=99.7 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=92.2 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=97.3 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=103 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=93.9 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=102 ms

--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5005ms
rtt min/avg/max/mdev = 92.175/98.089/103.404/4.067 ms
root@mininet-vm:/home/mininet#
```

Рисунок 8: Добавление значения дрожания задержки в интерфейс подключения

Добавление значения корреляции для джиттера и задержки в интерфейс подключения к эмулируемой глобальной сети

```
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root netem
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem delay 1
00ms 10ms 25%
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 20
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=107 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=94.1 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=103 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=107 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=94.8 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=98.9 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=92.8 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=102 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=110 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=107 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=103 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=96.4 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=93.1 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=98.2 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=103 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=110 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=109 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=101 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=105 ms
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=107 ms

--- 10.0.0.2 ping statistics ---
20 packets transmitted, 20 received, 0% packet loss, time 19041ms
rtt min/avg/max/mdev = 92.802/102.133/109.745/5.578 ms
root@mininet-vm:/home/mininet#
```

Распределение задержки в интерфейсе подключения к эмулируемой глобальной сети

```
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root netem
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem delay 1
00ms 20ms distribution normal
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 10
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=82.7 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=111 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=102 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=119 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=110 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=129 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=100 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=87.5 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=124 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=93.2 ms

--- 10.0.0.2 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 901ms
rtt min/avg/max/mdev = 82.720/105.785/129.487/14.727 ms
root@mininet-vm:/home/mininet#
```

Рисунок 10: Распределение задержки в интерфейсе подключения

```
mininet@mininet-vm: ~/work/lab_netem_i/simple-delay
#!/usr/bin/env python

"""
Simple experiment.
Output: ping.dat
"""

from mininet.net import Mininet
from mininet.node import Controller
from mininet.cli import CLI
from mininet.log import setLogLevel, info
import time

def emptyNet():

    "Create an empty network and add nodes to it."

    net = Mininet( controller=Controller,
                  waitConnected=True )

    info( '*** Adding controller\n' )
    net.addController( 'c0' )

    info( '*** Adding hosts\n' )
    h1 = net.addHost( 'h1', ip='10.0.0.1' )
    h2 = net.addHost( 'h2', ip='10.0.0.2' )

    info( '*** Adding switch\n' )
    s1 = net.addSwitch( 's1' )

    info( '*** Creating links\n' )
    net.addLink( h1, s1 )
    net.addLink( h2, s1 )

    info( '*** Starting network\n' )
    net.start()

    info( '*** Set delay\n' )
    h1.cmdPrint( 'tc qdisc add dev h1-eth0 root netem delay 100ms' )
    h2.cmdPrint( 'tc qdisc add dev h2-eth0 root netem delay 100ms' )

    time.sleep(10) # Wait 10 seconds

    info( '*** Ping\n' )
    h1.cmdPrint( 'ping -c 100', h2.IP(), '| grep "time=" | awk \'{print $5, $7}\'} | as
d -e \'s/time=//g\' -e \'s/icmp_seq=//g\' > ping.dat' )

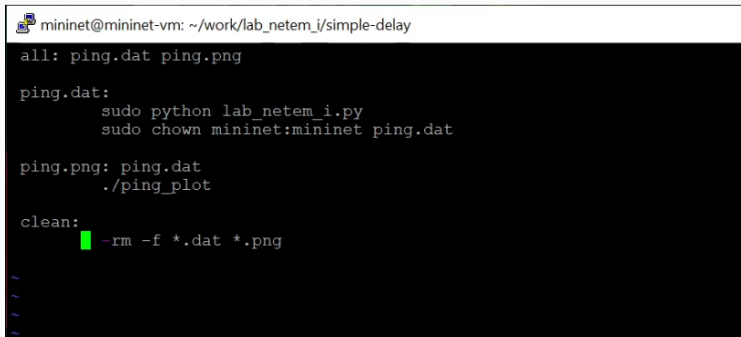
    info( '*** Stopping network' )
    net.stop()

if __name__ == '__main__':
    setLogLevel( 'info' )
    emptyNet()
~
```

 mininet@mininet-vm: ~/work/lab_netem_i/simple-delay

```
#!/usr/bin/gnuplot --persist  
  
set terminal png crop  
set output 'ping.png'  
set xlabel "Sequence number"  
set ylabel "Delay (ms)"  
set grid  
plot "ping.dat" with lines
```

Рисунок 12: скрипт для визуализации ping_plot



```
mininet@mininet-vm: ~/work/lab_netem_i/simple-delay
all: ping.dat ping.png

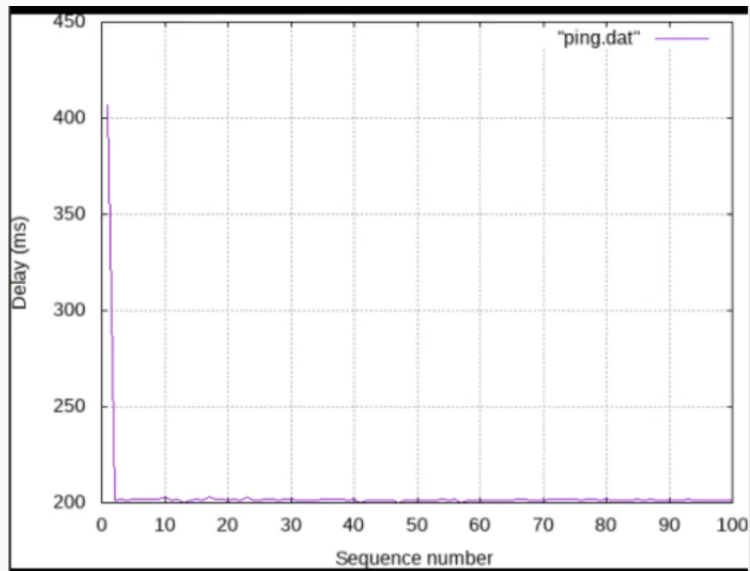
ping.dat:
    sudo python lab_netem_i.py
    sudo chown mininet:mininet ping.dat

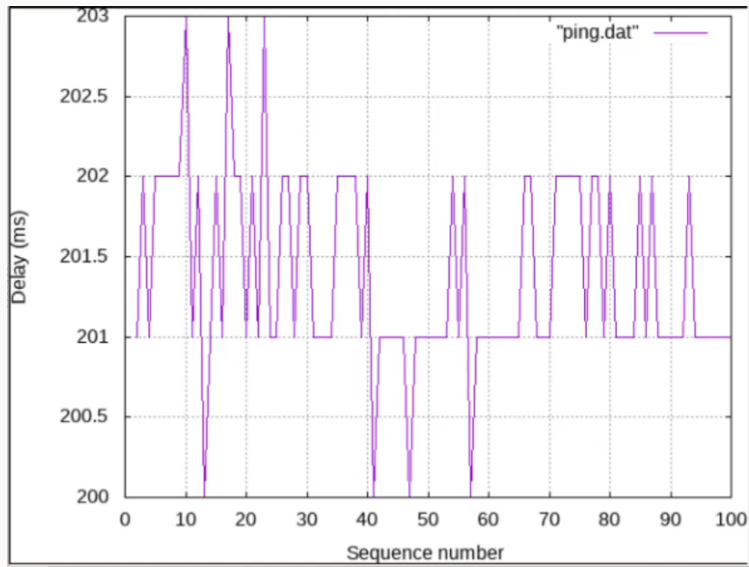
ping.png: ping.dat
    ./ping_plot

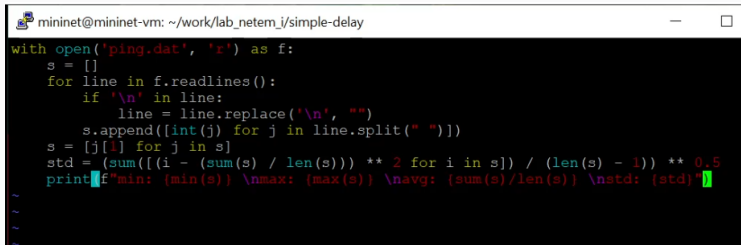
clean:
    -rm -f *.dat *.png

~
~
~
~
```

Рисунок 13: Makefile для управления процессом проведения эксперимента







```
mininet@mininet-vm: ~/work/lab_netem_i/simple-delay
with open('ping.dat', 'r') as f:
    s = []
    for line in f.readlines():
        if '\n' in line:
            line = line.replace('\n', '')
        s.append([int(j) for j in line.split(" ")])
    s = [j[1] for j in s]
    std = (sum([(i - (sum(s) / len(s))) ** 2 for i in s]) / (len(s) - 1)) ** 0.5
    print(f"min: {min(s)} \nmax: {max(s)} \navg: {sum(s)/len(s)} \nstd: {std}")
```

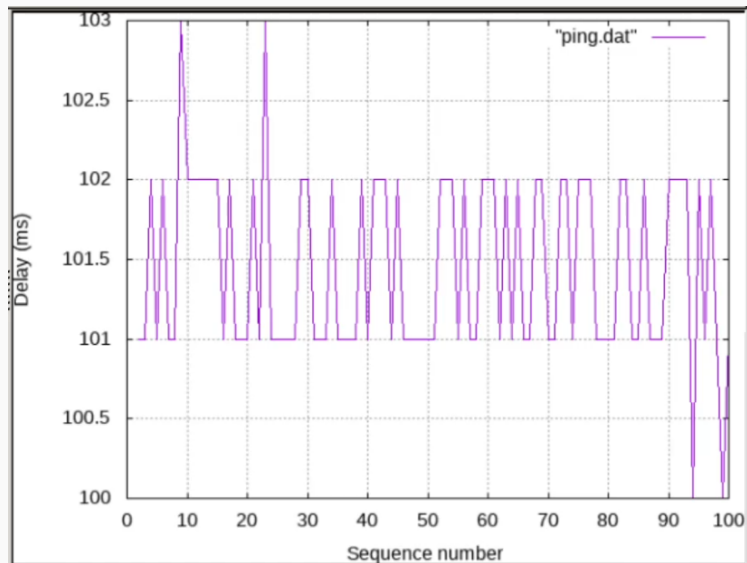
Рисунок 16: Скрипт rtt.py

```
mininet@mininet-vm: ~/work/lab_netem_i/simple-delay  
all: ping.dat ping.png  
  
ping.dat:  
    sudo python lab_netem_i.py  
    sudo chown mininet:mininet ping.dat  
    sudo python rtt.py  
  
ping.png: ping.dat  
    ./ping_plot  
  
clean:  
    -rm -f *.dat *.png
```

Рисунок 17: Makefile

Результат

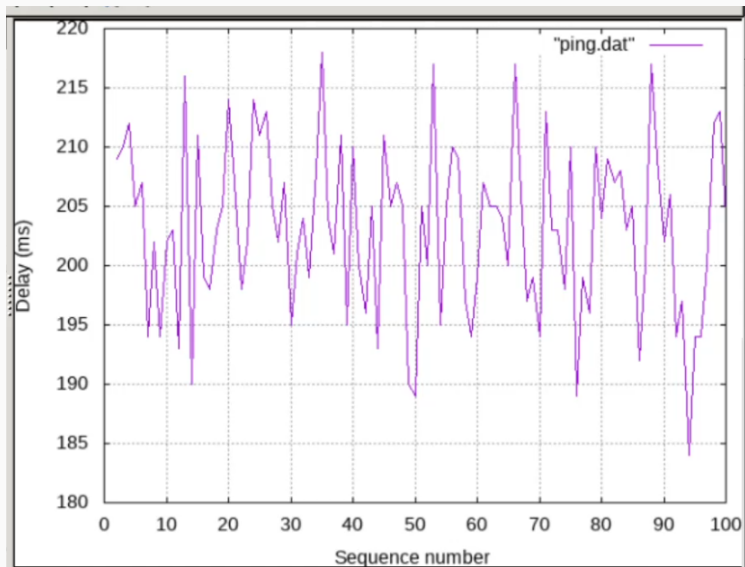
```
mininet@mininet-vm:~/work/lab_netem_i/simple-delay$ make
sudo python lab_netem_i.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Waiting for switches to connect
s1
*** Set delay
*** h1 : ('tc qdisc add dev h1-eth0 root netem delay 100ms',)
*** h2 : ('tc qdisc add dev h2-eth0 root netem delay 100ms',)
*** Ping
*** h1 : ('ping -c 100', '10.0.0.2', '| grep "time=" | awk \'{print $5, $7
\'s/time=//g\' -e \'s/icmp_seq=//g\' > ping.dat')
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
sudo chown mininet:mininet ping.dat
sudo python rtt.py
min: 200
max: 405
avg: 203.4
std: 20.37427571991485
./ping_plot
mininet@mininet-vm:~/work/lab_netem_i/simple-delay$
```



change-delay

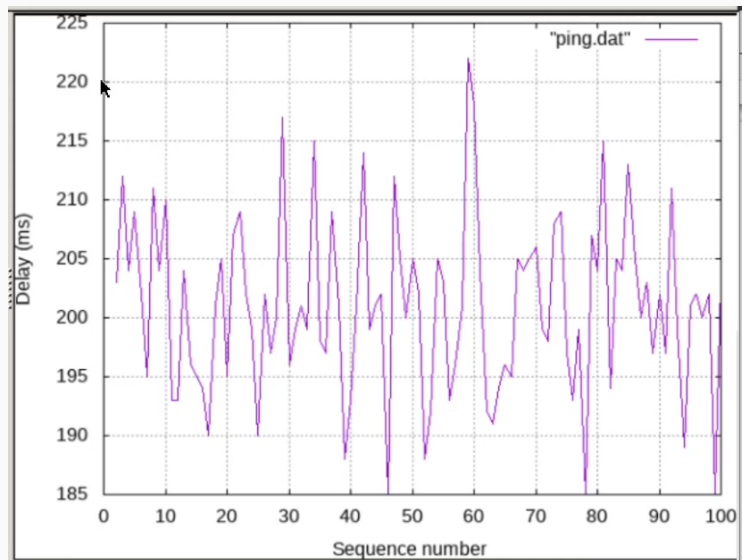
```
mininet@mininet-vm:~/work/lab_netem_i/change-delay$ ls
lab_netem_i.py  Makefile  ping_plot  rtt.py
mininet@mininet-vm:~/work/lab_netem_i/change-delay$ vim lab_netem_i.py
mininet@mininet-vm:~/work/lab_netem_i/change-delay$ vim Makefile
mininet@mininet-vm:~/work/lab_netem_i/change-delay$ make
sudo python lab_netem_i.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Waiting for switches to connect
s1
*** Set delay
*** h1 : ('tc qdisc add dev h1-eth0 root netem delay 100ms',)
*** h2 : ('tc qdisc add dev h2-eth0 root netem delay 100ms',)
*** Change delay
*** h1 : ('tc qdisc change dev h1-eth0 root netem delay 50ms',)
*** h2 : ('tc qdisc change dev h2-eth0 root netem delay 50ms',)
*** Ping
*** h1 : ('ping -c 100', '10.0.0.2', '| grep "time=" | awk \'{print $5, $7}\\' | sed -e
\'s/time=//g\' -e \'s/icmp_seq=//g\' > ping.dat')
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
sudo chown mininet:mininet ping.dat
sudo python rtt.py
min: 100
max: 206
avg: 102.5
std: 10.470256091723002
./ping_plot
```


jitter-delay



jitter-delay

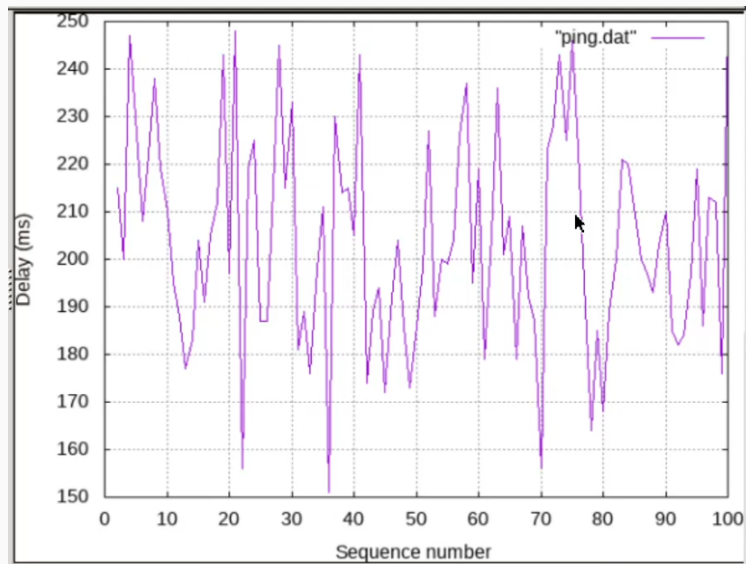
```
mininet@mininet-vm:~/work/lab_netem_i/jitter-delay$ make
sudo python lab_netem_i.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Waiting for switches to connect
s1
*** Set delay and jitter
*** h1 : ('tc qdisc add dev h1-eth0 root netem delay 100ms 10ms',)
*** h2 : ('tc qdisc add dev h2-eth0 root netem delay 100ms 10ms',)
*** Ping
*** h1 : ('ping -c 100', '10.0.0.2', '| grep "time=" | awk \'{print $5, $7}\\' | sed -e
\'s/time=//g\' -e \'s/icmp_seq=//g\' > ping.dat')
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
sudo chown mininet:mininet ping.dat
sudo python rtt.py
min: 184
max: 399
avg: 205.12
std: 20.8652628049395
./ping_plot
mininet@mininet-vm:~/work/lab_netem_i/jitter-delay$
```



correlation-delay

```
mininet@mininet-vm:~/work/lab_netem_i/correlation-delay$ make
sudo python lab_netem_i.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Waiting for switches to connect
s1
*** Set delay, jitter and correlation
*** h1 : ('tc qdisc add dev h1-eth0 root netem delay 100ms 10ms 25%',)
*** h2 : ('tc qdisc add dev h2-eth0 root netem delay 100ms 10ms 25%',)
*** Ping
*** h1 : ('ping -c 100', '10.0.0.2', '| grep "time=" | awk \'{print $5, $7}\\' | sed -
\'s/time=//g\' -e \'s/icmp_seq=//g\' > ping.dat')
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
sudo chown mininet:mininet ping.dat
sudo python rtt.py
min: 185
max: 417
avg: 203.14
std: 22.83538809600024
./ping_plot
```

distribution-delay



distribution-delay

```
mininet@mininet-vm:~/work/lab_netem_i/distribution-delay$ make
sudo python lab_netem_i.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Waiting for switches to connect
s1
*** Set delay and distribution
*** h1 : ('tc qdisc add dev h1-eth0 root netem delay 100ms 20ms distribution normal',)
*** h2 : ('tc qdisc add dev h2-eth0 root netem delay 100ms',)
*** Ping
*** h1 : ('ping -c 100', '10.0.0.2', '| grep "time=" | awk \'{print $5, $7}\\' | sed -e
\'s/time=//g\' -e \'s/icmp_seq=//g\' > ping.dat')
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
sudo chown mininet:mininet ping.dat
sudo python rtt.py
min: 151
max: 440
avg: 206.29
std: 32.42810073334109
./ping_plot
mininet@mininet-vm:~/work/lab_netem_i/distribution-delay$
```

В ходе выполнения лабораторной работы я познакомилась с NETEM – инструментом для тестирования производительности приложений в виртуальной сети, а также получила навыки проведения интерактивного и воспроизводимого экспериментов по измерению задержки и её дрожания (jitter) в моделируемой сети в среде Mininet.